



Chapter 1: Linear equations in Linear Algebra

Question 1:

Do the three lines $x_1 - 4x_2 = 1$, $2x_1 - x_2 = -3$ and $-x_1 - 3x_2 = 4$ have a common point of intersection? Explain.

Question 2:

Determine the value(s) of h such that the matrix is the augmented matrix of a consistent linear system:

$$\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 \\ -4 & h & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & h & -3 \\ -2 & 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & h \\ -6 & 9 & 5 \end{bmatrix}$$

Question 3:

Find an equation involving g , h , and k that makes this augmented matrix correspond to a consistent system:

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{bmatrix}$$

Question 4:

Find the elementary row operation that transforms the first matrix into the second, and then find the reverse row operation that transforms the second matrix into the first.

$$\begin{bmatrix} 0 & -2 & 5 \\ 1 & 4 & -7 \\ 3 & -1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & -7 \\ 0 & -2 & 5 \\ 6 & -2 & 12 \end{bmatrix}$$

Question 5:

Row reduce the matrices to reduced echelon form. Circle the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$$

Question 6:

Find the general solutions of the systems whose augmented matrices are given below:

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix}$$

Question 7:

Write a vector equation that is equivalent to the given system of equations:

$$\begin{aligned} x_2 + 5x_3 &= 0 \\ 4x_1 + 6x_2 - x_3 &= 0 \\ -x_1 + 3x_2 - 8x_3 &= 0 \end{aligned}$$

Question 8:

Determine if b is a linear combination of the vectors formed from the columns of the matrix A :

$$A = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 5 \\ -2 & 8 & -4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$$

Question 9:

Let $A = \begin{bmatrix} 2 & 0 & 6 \\ -1 & 8 & 5 \\ 1 & -2 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$ and let W be the set of all linear

combinations of the columns of A (or the $\text{Span} \{a_1, a_2, a_3\}$ where a_1, a_2, a_3 are the columns of A). Is b in W ?

Question 10:

Compute the products using (a) the definition, and (b) the row–vector rule for computing Ax .

$$A = \begin{bmatrix} 6 & 5 \\ -4 & -3 \\ 7 & 6 \end{bmatrix} \quad x = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

Question 11:

Describe all solutions of $Ax = 0$ in parametric vector form:

$$A = \begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \end{bmatrix}$$

Question 12:

Describe the solutions of the following system in parametric vector form:

$$\begin{aligned} x_1 + 3x_2 + x_3 &= 1 \\ -4x_1 - 9x_2 + 2x_3 &= -1 \\ -3x_2 - 6x_3 &= -3 \end{aligned}$$

Question 13:

Determine if the vectors are linearly independent:

a) $\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 7 \\ 2 \\ -6 \end{bmatrix}$ and $\begin{bmatrix} 9 \\ 4 \\ -8 \end{bmatrix}$

b) $\begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ 9 \end{bmatrix}$

c) $\begin{bmatrix} 0 & -8 & 5 \\ 3 & -7 & 4 \\ -1 & 5 & -4 \\ 1 & -3 & 2 \end{bmatrix}$

Question 14:

$$v_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -3 \\ 9 \\ -6 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 5 \\ -7 \\ h \end{bmatrix}$$

- a) For what values of h is v_3 in $\text{Span}\{v_1; v_2\}$?
- b) For what values of h is $\{v_1; v_2; v_3\}$ linearly dependent?

Question 15:

$$\text{Let } A = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}, \quad u = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} \text{ and } v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x) = A \cdot x$.

Find $T(u)$ and $T(v)$.

Question 16:

With T defined by $T(x) = Ax$, find a vector x whose image under T is b , and determine whether x is unique.

$$\text{a) } A = \begin{bmatrix} 1 & 0 & -2 \\ -2 & 1 & 6 \\ 3 & -2 & -5 \end{bmatrix}, \quad b = \begin{bmatrix} -1 \\ 7 \\ -3 \end{bmatrix}$$

$$\text{b) } A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

Question 17:

How many rows and columns must a matrix A have in order to define a mapping from $\mathbb{R}^4$ into $\mathbb{R}^5$ by the rule $T(x) = Ax$?

Question 18:

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $v_1 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$.

And let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps x into $x_1 v_1 + x_2 v_2$. Find a matrix A such that $T(x)$ is $A \cdot x$ for each x .

Question 19:

Find the standard matrix of T :

- a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(e_1) = (3, 1, 3, 1)$ and $T(e_2) = (-5, 2, 0, 0)$ where $e_1 = (1, 0)$ and $e_2 = (0, 1)$
- b) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(e_1) = (1, 3)$, $T(e_2) = (4, -7)$, $T(e_3) = (-5, 4)$ where e_1, e_2, e_3 are the columns of the identity matrix.
- c) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, is a horizontal shear transformation that leaves e_1 unchanged and maps e_2 into $e_2 + 3e_1$.

Question 20:

Determine if the specified linear transformation is one-to-one or onto. Justify each answer.

- a) $T(x_1, x_2, x_3, x_4) = (0, x_1 + x_2, x_2 + x_3, x_3 + x_4)$
- b) The transformation in Question 19 – b.